TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Young’s Modulus of the Material of a Wire by the Vernier Method

Stretching a long wire under known loads and reading its extension on a vernier scale — stress over strain, measured directly.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 3 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find Young’s modulus Y of a wire’s material. Within the elastic limit, stress is proportional to strain (Hooke’s law — Physics-I Topic 2), and

$$Y = \text{stress/strain} = (F/A) / (\Delta L/L) = F\,L / (A\,\Delta L) = 4\,M\,g\,L / (\pi\,d^2\,\Delta L)$$

for a load of mass M on a wire of length L, diameter d and extension ΔL. Extensions of a metal wire are fractions of a millimetre per kilogram, so a **long** wire is used (strain accumulates over length) and the extension is read on a **vernier scale**: a scale plate attached to the wire’s lower end moves past a fixed main scale, the vernier giving readings to 0.01 cm or finer. Extension per load step is plotted against load; the slope of the straight line through the origin is ΔL per unit force, and Y follows.

2. Apparatus and procedure

- **Apparatus:** the experimental wire (about 2 m) fixed at a rigid support, a companion dummy wire of the same material carrying the main scale (so support yielding and temperature drift cancel), scale pan, slotted masses, vernier arrangement, screw gauge, metre rule.
- Measure L (support to scale plate) and d (screw gauge, several places and orientations; the area depends on d^2, so this measurement dominates the error budget).
- Hang a small straightening load. Add masses in equal steps (e.g. 0.5 kg), reading the vernier at each step after the wire settles; then unload in the same steps, reading again. Loading and unloading readings that agree show the elastic limit was not crossed.
- Plot extension against load; take the slope from the straight line; compute Y.

3. Worked calculation (model data)

RESULT

Wire: L = 2.00 m; mean diameter d = 0.50 mm → radius 2.5×10^{-4} m; area $A = \pi(2.5 \times 10^{-4})^2 = 1.963 \times 10^{-7}$ m².

From the graph's slope: a load increase of 5.0 kg (force 49.0 N) produces $\Delta L = 2.50$ mm $= 2.50 \times 10^{-3}$ m.

$Y = F L/(A \Delta L) = (49.0 \times 2.00)/(1.963 \times 10^{-7} \times 2.50 \times 10^{-3}) = 98.0/(4.91 \times 10^{-10}) =$ **2.0×10^{11} Pa** — the accepted value for steel, as the model wire is steel.

Note the check the graph provides: stress at the largest load $= 49.0/1.963 \times 10^{-7} = 2.5 \times 10^{8}$ Pa, below steel's elastic limit — consistent with loading/unloading agreement.

4. Precautions

- Never exceed the elastic limit: a wire that does not return on unloading is ruined for the experiment (and its readings must be discarded).
- Wait after each load change — the extension creeps briefly as the wire settles and the temperature from handling fades.
- Measure the diameter with special care: Y depends on $1/d^2$, so a 2% error in d is a 4% error in Y.
- Add and remove loads gently, without jerking the pan or swinging it.
- The dummy-wire compensation only works if both wires are at the same temperature and the support is common — do not touch the wires during readings.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is a long wire used, and why a thin one?
2. A wire of L = 1.50 m, d = 0.40 mm extends 2.34 mm under 4.0 kg. Compute Y.
3. Why are readings taken both while loading and while unloading?
4. What is the purpose of the dummy (companion) wire?
5. Why does the diameter measurement dominate the error in Y?
6. The extension–load graph bends near the top loads. What has happened, and what must be done with those points?

Answer key

1. Extension $\Delta L = FL/(AY)$ grows with L and shrinks with area: a long, thin wire stretches measurably under safe loads, turning a sub-millimetre effect into a readable scale movement while the stress

stays below the elastic limit.

2. $A = \pi(2.0 \times 10^{-4})^2 = 1.257 \times 10^{-7}$ m²; F = 39.2 N; $Y = (39.2 \times 1.50)/(1.257 \times 10^{-7} \times 2.34 \times 10^{-3}) = 58.8/(2.941 \times 10^{-10}) = 2.0 \times 10^{11}$ Pa — steel's accepted value.
3. To prove the wire stayed elastic. If it did, unloading retraces loading exactly; any permanent set shows as a zero shift and warns that the elastic limit was crossed and the data are invalid beyond that load.
4. It carries the main scale and experiences the same support movement and ambient temperature changes as the experimental wire, so those common disturbances cancel in the relative (vernier) reading — only the experimental wire's true extension is measured.
5. Because $Y \propto 1/d^2$: the relative error in Y from the diameter is twice the relative error in d, and d is the smallest quantity in the experiment (a fraction of a millimetre measured to hundredths). It is the precision bottleneck.
6. The wire has passed its elastic limit (proportionality has failed — Hooke's law no longer holds). The bent points must be excluded, the maximum load reduced, and ideally a fresh wire used, since the old one may have taken a permanent set.
